Name: Tejas Rao
Instructor: [redacted]
Assignment: **eometry n y or**
Course: DATA239 – Geometry for Computer Vision
Date: November 30, 2025

Abstract

This work, after briefly highlighting how the course changed my view of geometry, explores its role across three application areas. The first is in Data Science and Machine Learning applied to Biology, where I discuss how some transformations we studies power data augmentation, define feature spaces, and govern classification algorithms. Even in molecular biology, constraints like Euclidean distance inequalities underlie breakthroughs such as AlphaFold 2. I also discuss my interest in Visual Arts (Film and Photography), where projective geometry defines image formation. I discuss geometric phenomena like symmetries, and vanishing points being appropriated as compositional tools by Filmmakers, as well as a brief section on historical perspective systems. Lastly, I discuss some geometry in Architecture and Everyday Structures. Here, I draw attention to instances of isometries, affine and projective distortions, as well as property-invariance under projective transformations in local, real-world contexts.

Introduction

This course expanded and formalized the geometric intuition I carried prior to studying Euclidean, Affine, and Projective Geometry in a structured manner. Earlier, I could vaguely sense connections between many artifacts but I lacked the formal lens to articulate why. For example, I knew that shapes like circles and ellipses were related, though not precisely how. Through affine and projective tools, I now understand that an ellipse can be viewed as the parallel projection of a circle when the source and image planes are skewed; the angle between these planes controls the eccentricity of the ellipse. These precise formulations allowed me to refine intuition into verifiable statements. Another example is how the 'Kleinian' view of geometry corroborated the my implicit notion of 'other' geometries (I had encountered non-euclidean geometry before), in a formalised, systematic way which made it easy to intuit through the 'other' geometries. Further, I was also unfamiliar with projective geometry before this course. Concepts such as the horizon and vanishing points previously registered only as artistic constructions in photography and film. Now, I recognize them as mathematical 'consequences' of central projection and points "at infinity" becoming finite under transformation. This shift from informal visual reasoning to rigorous geometric interpretation has changed how I view images, camera models, and architectural perception on some level.
Another major takeaway is how central linear algebra is to vision and graphics. The course continuously demonstrated that transformations like Euclidean isometries, affine mappings,

or other projective transformations are maps represented with matrices. This clarified how modern computer vision pipelines encode geometry at their core: through coordinate transformations, feature manipulations, and constraints expressed in linear algebraic terms.
This project examines how geometric frameworks introduced in euclidean, affine, and projective geometry manifest in three domains that structure both my academic work and personal interests: data science and machine learning in biology, visual arts, and local architecture. In biological ML applications, I analyze how transformations such as translations, rotations, scalings, and other affine mappings support data augmentation, define feature spaces, and govern geometric classifications—illustrated by techniques like SMOTE and geometric constraints used in protein structure prediction. In the visual arts, I reinterpret composition, symmetry, and perspective in photography and cinema through projective models of image formation. Here phenomena like Symmetries, vanishing points, central projection, and affine distortions are harnessed to be active tools of visual storytelling. The final section demonstrates how similar principles/phenomena appear in the built environment, visible through isometries in tilings, affine shadow projections, and cross-ratio preservation across viewpoints. Together, these case studies highlight a coherent viewpoint: geometry is not merely a study of ideal shapes, but rather a useful language for how information, images, and spaces are represented and transformed in the real world.

Geometry in my field of study: Data Science and ML Applications in Biology

The geometries we have covered in this course has numerous applications in Data Science, which have been repeatedly used in the Biological Sciences. In tasks pertinent to Computer Vision, various tools from Geometry are used for the so-called 'feature engineering'. For reference, features refer to what a model receives as an input and uses to come to a prediction. In Computer Vision tasks, the features, at least for the most part, are the pixels of an image. Often, especially for highly-specialized tasks, there isn't enough data for a model to capture trends that help in accurate prediction on unseen data. To promote model's ability to generalize to unseen data, (or more broadly, to situations that aren't in the training set), a technique adopted is Data Augmentation. Here, we 'synthesize' new images by applying a set of Transformations to the training set, thereby enhancing the apparent number of datapoints the model trains on. These transformations are usually Euclidean (Translations, Rotations, Reflections) or Affine (Scaling, Shearing, Distortions). However, there are other (non-linear) methods that don't fall into the classes of transformations we studied but are important for data augmentation (E.g., adding noise, blurring, adversarial perturbations etc). A few examples are given in Figure 1:

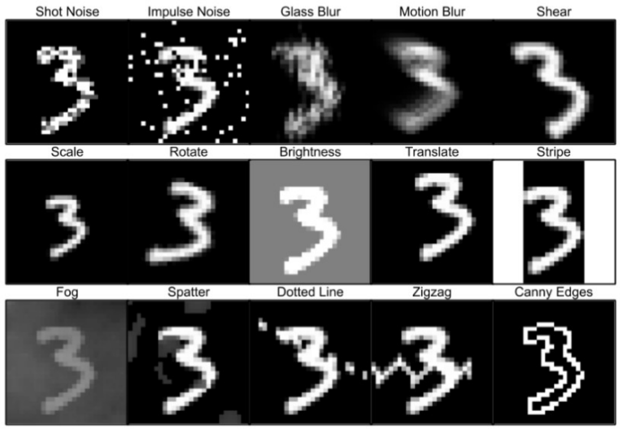


Figure 1: Examples for Data Augementation. Image Source from a paper on computer vision benchmarking: [5]

Sticking to Data Augmentation, an interesting technique is creating data that looks real, but is not actually real. A really common method for synthetic data generation is the 'Synthetic Minority Oversampling Technique', or SMOTE. If we consider more abstract notions of higher-dimensional geometry, it can be construed as a fundamentally geometric technique. Like most classical machine learning techniques, it operates in the 'feature space' (usually $\mathbb{R}^n$). The algorithm randomly chooses a 'real' datapoint x and considers randomly one of it's real neighbours x'. Then, a new datapoint $\tilde{x}$ is synthesised by choosing a point on the displacement vector between x and x', i.e., $\alpha \times (x' - x)$ where $\alpha \in (0, 1)$. Essentially, the way it generates synthetic data is governed by this equation:

$$\tilde{x} = x + \alpha \times (x' - x)$$

This formulation of SMOTE shows us how the algorithm essentially performs linear interpolations in the feature space (as shown in Figure 2.

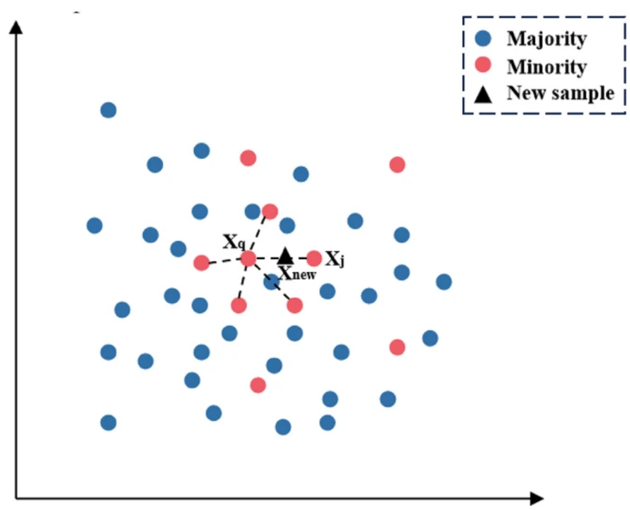


Figure 2: Inforgraphic illustrating SMOTE. Source: [4]

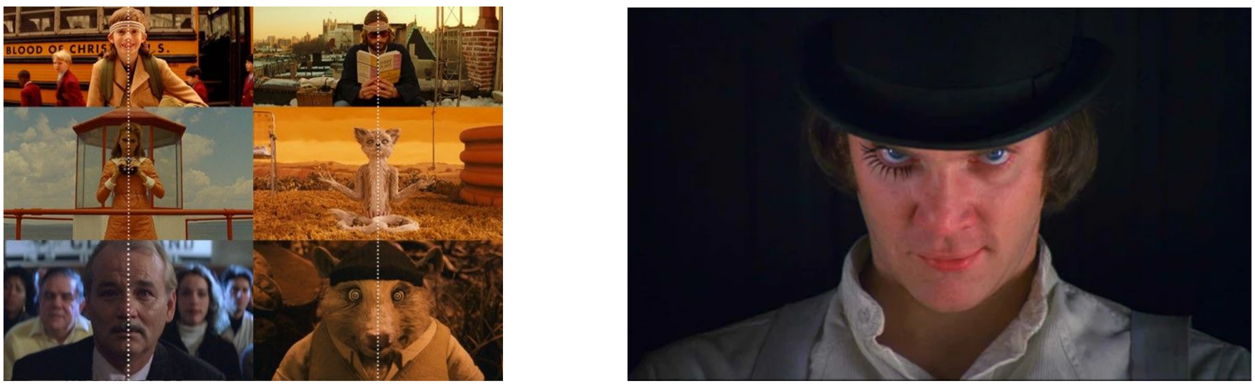
In Biology, such data augmentation methods are prevalent in Machine Learning implementations in fields like Neuroimaging, Microscopy, Cell Painting and so forth. This is because often, there aren't large datasets for the specific phenomena being studied, which can result in jagged loss landscapes, and reduced statistical power.
Many ML techniques for classification can also be interpreted as a geometric problem. At their core, they seek to determine a decision boundary that partitions the feature space into disjoint regions corresponding to different classes in the training set. Each data point, represented as a feature vector in this high-dimensional space, is classified by assessing its relative position with respect to this boundary. A fascinating example is the Support Vector Machine algorithm (SVM). Here, the separating surface is a hyperplane embedded in a 'potentially' infinite-dimensional Hilbert space via kernel mappings. The objective is to maximize the geometric margin between classes, yielding a decision rule that is both robust and geometrically interpretable.

Further geometric perspective extends to other cutting-edge biological applications. In the Nobel Prize–winning AlphaFold 2 framework for protein structure prediction, geometric constraints from Euclidean space fundamentally guide the learning process [1]. Specifically, predicted inter-residue distances must satisfy the triangle inequality, a core rule of Euclidean geometry ensuring that the distance between any two amino acids cannot exceed the sum of their distances to a third. By embedding these structural priors directly into model design and loss functions, AlphaFold 2 enforces physically realizable protein conformations.

Geometry in Personal Interests: Visual Arts

Geometry is fundamentally embedded in the visual arts, like photography, cinematography, painting, sculpture, and beyond. At a foundational (and obvious) level, for cinema and photography, the very act of image formation in cameras is governed by projective geometry: a three-dimensional scene is mapped to a two-dimensional image plane through central projection.
On a higher level, composition is an art often involving geometrically informed heuristics such as the golden ratio, leading lines (often incorporated with the vanishing points) and the rule of thirds. Compositional tools such as symmetry (an isometry: reflection of one part of an object about a line is equal to the complementary part of the object) are invoked not merely for aesthetic pleasantness, but additionally as mechanisms for conveying affective or narrative intent. The filmmaker Wes Anderson, a formalist, has cultivated a recognisable artistic signature through the deliberate deployment of axial symmetries to impose order, whimsy, and stylised control over the frame, as in Figure 3 (a). In contrast, Stanley Kubrick's practice demonstrates both the expressive power of symmetry and the emotional tension induced through 'near-symmetry'. For example, the purposeful asymmetry in the face of Alex DeLarge in *A Clockwork Orange* visually signals the character's behavioural and moral deviation despite the otherwise "normal" appearance (refer Figure 3 (b)). Likewise, in *The Shining*, Kubrick repeatedly invokes "almost symmetry" to elicit unease—famously exemplified by the appearance of the Grady twins in the tricycle corridor sequence, where rigid architectural regularity is haunted by a subtle, uncanny misalignment in human form.

Figure 4: A frame from the Tricycle scene, where Kubrick's use of projective frames and one-point perspective for suspense. Source *The Shining*



(a) (b)

Figure 3: (a) Wes Anderson's use of symmetry for visual storytelling (source: Pinterest). (b) Alex DeLarge as an example of Kubrick's 'near-symmetry'. (source: *Clockwork Orange*

Kubrick also exploits perspective as an artistic fashion (famously, the one-point perspective), often combining it with his use of symmetry. In the same tricycle scene in *The Shining*, Kubrick positions the camera very low with a relatively narrow field of view, so the projective mapping from the three-dimensional corridor (object space) onto the two-dimensional image (picture plane) forces parallel lines to converge steeply toward a vanishing point. This low center of perspective increases occlusion and 'compresses' visible information with distance, meaning that anything just beyond each corner drops out of view more abruptly. This makes the space feel more mysterious and threatening and the viewer anticipates danger at the limits of perspective, which combines well with the 'symmetry to elicit discomfort' notion described above.

Any discussion of projective geometry in visual art is incomplete without mention of the Renaissance. As an analysis suggests, the Renaissance adoption of vanishing points, implicitly adopts the projective model of geometry, way before the rigorous formalisation by contributors like Desargues and Poncelet [2]. Alberti's perspective construction, evident in works like *De pictura*, presents the painting as a "transparent window" through which visual rays intersect a plane, making the picture a projection from a fixed centre [2, 3]. This translates directly to fundamental projective principles: straight lines map to straight lines, incidence is preserved, and families of parallel lines acquire finite intersection points in the form of vanishing points. This is also reflected in Masaccio's work, for instance in *Holy Trinity*. Here, the orthogonals receding from the viewer converge to a single vanishing point, reflect that the affine directions of real space are completed by points at infinity in the image. From a modern standpoint, this transformation is naturally expressed using homogeneous coordinates, where points differing by a non-zero scalar represent the same direction, and parallelism is captured by a class of ideal points.

Geometry in Local Architecture

Isometries and Projective Geometry: The ceiling of the Chennai Central Railway Station consists of many (idealized) square tessellations that are clipped by perspective (Figure 5 (a)). Each 'square unit' of area say l^2 is translated both vertically and horizontally by a side length l to form the grid seen above. Another way to construe this, is to observe that the square unit is reflected about its side n times to form longer rows of squares (which is eventually a rectangle). Then these rows are reflected about their longer side (of length $n \times l$) to make a grid. However, in the image this ideal behavior is not immediately apparent. This is because the Perspective projection introduces projective distortions, such that Squares no longer appear congruent, Parallel lines converge to vanishing points and Distances and angles not preserved. A similar example with parallelograms is visible in an image I had captured (Figure 5 (b)). It is relevant to this section as we observe the overlapping effects of reflections (isometries) as well as vanishing points.

(a)

(b)

Figure 5: (a) An example of isometries in a 2D plane that is observed 'projectively'. (b) An image showing isometries converging to infinity

Library Building Circles: This wall features perfect Euclidean circles of different radii embedded in a plane. Take the large bottom–left circle be C_0 and centered at the origin with radius R. Any other circle in the flat image, with center $c_i = (a_i, b_i)^T$ and radius R_i, can be obtained via the affine transformation: $T_i(x) = S_i\, x + t_i$,, where:

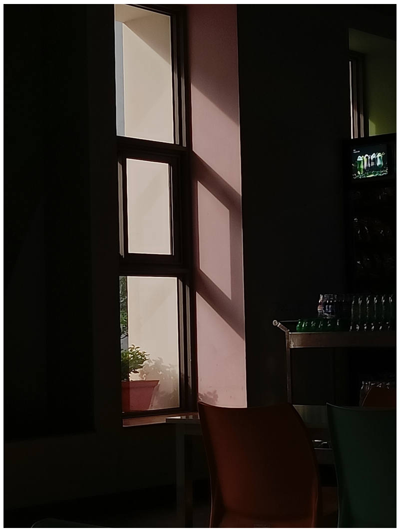
$$S_i = \begin{pmatrix} s_i & 0 \\ 0 & s_i \end{pmatrix}, \qquad s_i = \frac{R_i}{R}, \qquad t_i = \begin{pmatrix} a_i \\ b_i \end{pmatrix}.$$

Figure 6: Viewing circles congruent under the operation of scaling and translation transformations

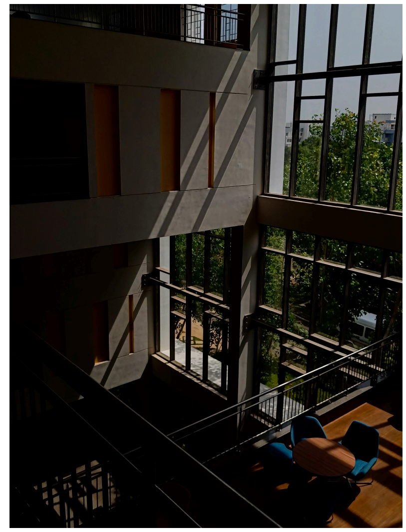
For the smaller upper circles we have $R_i < R$, meaning

$$s_i = \frac{R_i}{R} < 1 \Rightarrow \det(S_i) = s_i^2 < 1.$$

Parallel Projections in Action in Krea: This is a similar observation to those mentioned in one of my geometry portfolio assignments. We can model this phenomena seen in Figure 7 subimage (a) as parallel projections: the sun's rays are parallel lines, with π_1 being the vertical wall containing the window frame and π_2 being the interior wall on which the shadow is cast. Another instance is visible in Krea's New Academic Block (Figure 7 subimage (b)). In this space as well, the sunlight can be modeled as a family of parallel rays, producing an affine parallel projection from the plane of the window frame to interior architectural surfaces such as the stairwell walls and floor. The window grid — made of parallel vertical and horizontal segments in π_1 — projects as parallel but skewed segments on π_2 creating the long diagonal shadow bands visible in the image.



(a)

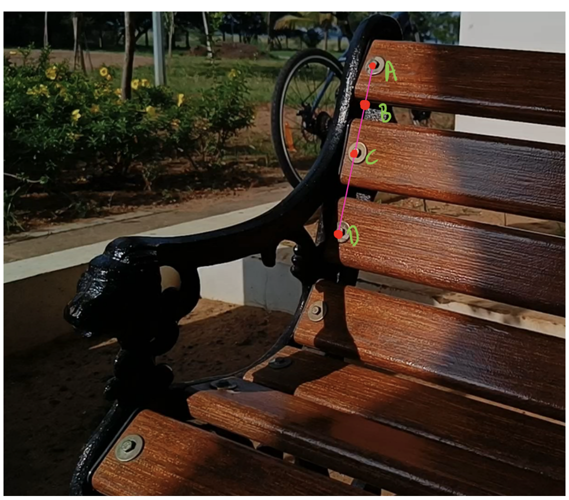


(b)

Figure 7: *a)* An example of a parallel projection created by the Krea Dining Hall Windows. *b)* A similar observation in Krea's New Academic Block

In both cases, we notice that parallel straight lines map to parallel straight lines, and distances & angles are not preserved (i.e., rectangles map to parallelograms).

Cross Ratio Invariance in Architectural Components: We see that the cross-ratio is a invariant property under projective transformations. It then follows that for 2 images of the same object taken from different perspectives, the cross ratio for 4 collinear points should be the same. We illustrate this using Figure 8.



(a)

(b)

Figure 8: Images of a bench to illustrate cross-ratio invariance.

For both images, we mark the collinear points A, B, C and D for cross ratio calculation. The distances between points were calculated using the ruler feature in a drawing software. For subimage (a), the measurements were $AB = 0.6$ units, $BC = 0.8$ units, $CD = 1.25$ units. The cross ratio λ_1 for sub-image (a) is given by

$$\lambda_1 = [AB : CD] = \frac{AC \cdot BD}{BC \cdot AD} = \frac{1.4 \times 2.05}{0.8 \times 2.65} \implies \lambda_1 \approx 1.35$$

Similarly for subimage (b), the measurements were $AB = 0.45$ units, $BC = 0.65$ units, $CD = 1$ unit. The cross ratio λ_2 for subimage (b) is given by

$$\lambda_2 = [AB : CD] = \frac{AC \cdot BD}{BC \cdot AD} = \frac{1.1 \times 1.65}{0.65 \times 2.1} \implies \lambda_2 \approx 1.33$$

Thus, $\lambda_1 \approx \lambda_2$, which shows the invariance of cross-ratios under projective transformations. We see that a cross ratio of ≈ 1.3 is recovered from both perspectives (This figure is only approximate account due to measuring inconsistencies, distortions and other errors).

Conclusion

The three application areas show that geometry consistently governs how objects are represented, compared, and transformed. In data science and biology-focused ML, operations like interpolation, classification, and structural validation are geometric in feature or Euclidean spaces. In visual arts, perspective systems and symmetries follow projective rules that dictate how a scene is mapped to an image. In architecture, everyday perception is shaped by affine and projective distortions, with measurable invariants such as the cross-ratio persisting across viewpoints.

The most surprising insight for me was recognizing that techniques I routinely apply in ML are geometric processes written in numerical form. This shifted my view of geometry from a subject concerned with idealized shapes to a framework that underlies practical computation, visual representation, and how we interpret the physical abstract numerical environments, implying that Geometry is a 'synthesis' of sorts, with the same principles operating across data, images, and space.

References

[1] Justas Dauparas and Fabian Fuchs. Alphafold 2 & equivariance. https://fabianfuchsml.github.io/alphafold2/, December 2020. Blog post.

[2] Graziano Gentili, Luisa Simonutti, and Daniele C. Struppa. The mathematics of painting: the birth of projective geometry in the italian renaissance. *arXiv preprint*, 2022.

[3] Jonathan Janson. The history of perspective, n.d. Accessed 30 November 2025.

[4] Ying Li, Yali Yang, Peihua Song, Lian Duan, and Rui Ren. An improved smote algorithm for enhanced imbalanced data classification by expanding sample generation space. *Scientific Reports*, 15(1):23521, 2025.

[5] Norman Mu and Justin Gilmer. Mnist-c: A robustness benchmark for computer vision. *arXiv preprint arXiv:1906.02337*, 2019.